01 Fuel Pump Relay Output
02 Idle valve close control
03 Injector 5 (M50TU): injector 1 (M60/S50/M50)
04 Injector 6 (M50TU): injector 4 (M60): injector 3 (S50/M50)
05 Injector 4 (M50TU): injector 6 (M60): injector 2 (S50/M50)
06 Power/Chassis GND ground (Injector/output/ignition return)
07 VANOS solenoid (M50TU) / injector 7 (M60)
08 Check Engine Light Output
09 Unspecified OEM terminal; independent pass-through
10 Unspecified OEM terminal; independent pass-through
11 Throttle-position output to transmission / traction control (variant dependent)
12 Universal analog input. Suggested TPS 1 sensor input (M50); oxygen sensor 2 input (M60/S50)
13 Oxygen sensor 1 input (M50TU/M60/S50) / MAF burn-off (M50)
14 GNDA Analog/Sensor Ground (MAF return; JP14 to ECU GNDA)
15 Ignition monitor / return (do not assume chassis ground)
16 Crank sensor input (M50TU/M60/S50) / cam sensor negative (M50)
17 Cam sensor input (M50TU/M60/S50) / fuel-consumption output (M50)
18 Auxiliary circuit (not specified for M3x in reference)
19 Secondary air pump control (M60/S50: verify equipment)
20 Unspecified OEM terminal; independent pass-through
21 Unspecified OEM terminal; independent pass-through
22 Ignition primary output, driver channel 7 (M60: coil 7)
23 Ignition primary output, driver channel 4 (M50TU: coil 4; M60: coil 6; S50: coil 2; M50: coil 2)
24 Ignition primary output, driver channel 6 (M50TU: coil 6; M60: coil 4; S50: coil 3; M50: coil 3)
25 Ignition primary output, driver channel 5 (M50TU: coil 5; M60: coil 1; S50: coil 1; M50: coil 1)
26 12V Battery Supply (constant, terminal 30)
27 Main Relay Control Output
28 Power/Chassis GND ground (electronics/shields: connected to chassis in M50TU/M60 wiring: JP28 …
29 Idle valve open control
30 Unspecified OEM terminal; independent pass-through

J3A

1	OUT_FUEL_PUMP
2	OUT_IDLE_CLOSE
3	OUT_INJ_P03
4	OUT_INJ_P04
5	OUT_INJ_P05
6	GND
7	OUT_INJ_VVT_P07
8	OUT_CHECK_ENGINE
9	OEM_09
10	OEM_10
11	OEM_11
12	OEM_12
13	OEM_13
14	GNDA_MAF_P14
15	OEM_15
16	IN_CRANK_P16
17	IN_CAM_P17
18	OEM_18
19	OEM_19
20	OEM_20
21	OEM_21
22	OUT_HV_IGN_7
23	OUT_HV_IGN_4
24	OUT_HV_IGN_6
25	OUT_HV_IGN_5
26	IN_CONSTANT_POWER
27	OUT_MAIN_RELAY
28	GND_ELECTRONICS_P28
29	OUT_IDLE_OPEN
30	OEM_30

31 Injector 3 (M50TU): injector 5 (M60/S50/M50)
32 Injector 2 (M50TU): injector 8 (M60): injector 6 (S50/M50)
33 Injector 1 (M50TU): injector 3 (M60): injector 4 (S50/M50)
34 Power/Chassis GND ground (Injector/output/ignition return)
35 Injector 2 output (M60)
36 EVAP canister purge valve control
37 Oxygen sensor heater relay control (M60/S50/M50: separate from pin 38)
38 Oxygen sensor heater relay control (M50TU: separate from pin 37)
39 GNDA Analog/Sensor Ground (oxygen sensor 2 return, M60/S50: JP39 to ECU GNDA)
40 GNDA Analog/Sensor Ground (oxygen sensor 1 return, M50TU/M60/S50: JP40 to ECU GNDA)
41 MAF Sensor Input
42 Vehicle speed input (M50TU/M60/S50)
43 Crank sensor return (M50TU/M60/S50): sensor ground on M50 reference
44 Cam sensor return (M50TU/M60/S50) / cam sensor positive (M50)
45 Ignition-monitor shield / return (verify harness)
46 Fuel consumption output (TI)
47 Tachometer Output (TD: M50TU/M60/S50)
48 A/C compressor relay control
49 Ignition primary output, driver channel 8 (M60: coil 2)
50 Ignition primary output, driver channel 1 (M50TU: coil 1; M60: coil 3; S50: coil 4; M50: coil 4)
51 Ignition primary output, driver channel 2 (M50TU: coil 2; M60: coil 8; S50: coil 6; M50: coil 6)
52 Ignition primary output, driver channel 3 (M50TU: coil 3; M60: coil 5; S50: coil 5; M50: coil 5)
53 Unspecified OEM terminal; independent pass-through
54 12V supply from main relay (terminal 87)
55 Power/Chassis GND ground (Injector/output/ignition return)
56 Ignition Switch / Battery Voltage Analog Input (terminal 15)
57 Transmission Ignition-timing Intervention (M50TU)
58 Unspecified OEM terminal; independent pass-through
59 5V Sensor Supply (TPS)
60 Diagnostic programming-voltage circuit (verify protocol)

J3B

31	OUT_INJ_P31
32	OUT_INJ_P32
33	OUT_INJ_P33
34	GND
35	OUT_INJ_P35
36	OUT_EVAP
37	OUT_O2_RELAY_P37
38	OUT_O2_RELAY_P38
39	GNDA_O2_P39
40	GNDA_O2_P40
41	IN_MAF
42	IN_VSS_P42
43	CRANK_RETURN_P43
44	CAM_RETURN_P44
45	GND_IGN_SHIELD_P45
46	OUT_FUEL_CONSUMPTION
47	OUT_TACH_P47
48	OUT_AC_RELAY
49	OUT_HV_IGN_8
50	OUT_HV_IGN_1
51	OUT_HV_IGN_2
52	OUT_HV_IGN_3
53	OEM_53
54	12V_FROM_MAIN_RELAY
55	GND
56	IN_VIGN
57	OEM_57
58	OEM_58
59	+5VAS
60	OEM_60

61 Unspecified OEM terminal; independent pass-through
62 Transmission / traction-control signal (variant dependent)
63 Unspecified OEM terminal; independent pass-through
64 A/C request (M50TU/M60/S50) / transmission timing intervention (M50)
65 A/C pressure input (M50TU/M60/S50) / transmission P/N (M50)
66 Immobilizer / drive-away protection input (variant dependent)
67 Knock sensor, cylinders 7-8 (M60) / crank negative (M50)
68 Knock sensor, cylinders 5-6 (M60/S50) / crank positive (M50)
69 Knock sensor, cylinders 4-6 (M50TU) or 3-4 (M60/S50)
70 Knock sensor, cylinders 1-3 (M50TU) or 1-2 (M60/S50) / oxygen signal (M50)
71 Knock-sensor shield/return (M50TU/M60/S50) / oxygen sensor return (M50): JP71 selects GND or G…
72 Unspecified OEM terminal; independent pass-through
73 Universal analog input. Suggested TPS 1 sensor input (M50TU/M60/S50); VSS on M50
74 Tachometer output (M50 reference: otherwise unspecified)
75 Unspecified OEM terminal; independent pass-through
76 Unspecified OEM terminal; independent pass-through
77 IAT Sensor Input
78 ECT CLT Coolant Sensor Input
79 Unspecified OEM terminal; independent pass-through
80 Unspecified OEM terminal; independent pass-through
81 Transmission P/N (M50TU): headlights-on signal (M60): verify variant
82 ABS/traction-control interface (verify harness)
83 ABS ignition intervention (verify harness)
84 Shield / return (M60/S50): independent contact
85 CAN bus Low (M60/S50) / A/C pressure input (M50)
86 CAN bus High (M60/S50) / auxiliary fan signal (M50)
87 Diagnostic RXD circuit
88 Diagnostic TXD circuit

J3C

61	OEM_61
62	OEM_62
63	OEM_63
64	IN_AC_P64
65	IN_AC_P65
66	IN_IMMOBILIZER
67	IN_KNOCK_P67
68	IN_KNOCK_P68
69	IN_KNOCK_P69
70	IN_KNOCK_P70
71	RETURN_P71
72	OEM_72
73	IN_TPS_P73
74	OEM_74
75	OEM_75
76	OEM_76
77	IN_IAT
78	IN_CLT
79	OEM_79
80	OEM_80
81	OEM_81
82	OEM_82
83	OEM_83
84	GND_CAN_SHIELD_P84
85	OEM_85
86	OEM_86
87	OEM_87
88	OEM_88

BMW 88 / Superseal 60+34 — eight ignition power stages
Scope: Bosch M3.1/M3.3/M3.3.1. NOT Siemens MS40/MS41. Cylinder order differs by engine.
J3 pins 6/34/55 are power GND. All other OEM pins are independent. All JPs are OPEN.
GNDA is ECU sensor ground on 30C. JP14/39/40 join sensor returns to GNDA.
JP28/45/84 join chassis/shields to GND. JP71: 1=GNDA. 2=OEM71. 3=GND; bridge one side only.
Do not close links on signal variants. Pins 15/44 are deliberately not ground-jumpered.
Q1–Q8: 22 ohm series gate resistance; 100k input pulldown, same topology as Ford adapter.
R17–R24: 0 ohm 2512 bypasses. DNP by default. Smart coils: remove Qn, fit R(n+16). NEVER BOTH.
Use 5 V ignition commands and ECU-controlled dwell. Channel number is not cylinder number.
No dwell timeout/current limiter. Load and thermal validation required for assembled hardware.
A=J1.1…34. B=J1.35…60. C=J2.1…34. Pinouts: rusefi.com/docs/pinouts/bmw88/ and /bmw88-adapter/.

R17 0 OUT_HV_IGN_1 IN_IGN_1
Q1 ISL9V3040D3S GATE_1 R1 22 IN_IGN_1 R9 100k GND
CH1: OEM 50; set cylinder order in ECU

R18 0 OUT_HV_IGN_2 IN_IGN_2
Q2 ISL9V3040D3S GATE_2 R2 22 IN_IGN_2 R10 100k GND
CH2: OEM 51; set cylinder order in ECU

R19 0 OUT_HV_IGN_3 IN_IGN_3
Q3 ISL9V3040D3S GATE_3 R3 22 IN_IGN_3 R11 100k GND
CH3: OEM 52; set cylinder order in ECU

R20 0 OUT_HV_IGN_4 IN_IGN_4
Q4 ISL9V3040D3S GATE_4 R4 22 IN_IGN_4 R12 100k GND
CH4: OEM 23; set cylinder order in ECU

R21 0 OUT_HV_IGN_5 IN_IGN_5
Q5 ISL9V3040D3S GATE_5 R5 22 IN_IGN_5 R13 100k GND
CH5: OEM 25; set cylinder order in ECU

R22 0 OUT_HV_IGN_6 IN_IGN_6
Q6 ISL9V3040D3S GATE_6 R6 22 IN_IGN_6 R14 100k GND
CH6: OEM 24; set cylinder order in ECU

R23 0 OUT_HV_IGN_7 IN_IGN_7
Q7 ISL9V3040D3S GATE_7 R7 22 IN_IGN_7 R15 100k GND
CH7: OEM 22; set cylinder order in ECU

R24 0 OUT_HV_IGN_8 IN_IGN_8
Q8 ISL9V3040D3S GATE_8 R8 22 IN_IGN_8 R16 100k GND
CH8: OEM 49; set cylinder order in ECU

JP14 GNDA_MAF_P14 GNDA
JP39 GNDA_O2_P39 GNDA
JP40 GNDA_O2_P40 GNDA
JP45 GND_IGN_SHIELD_P45 GND
JP71 1 GNDA 2 RETURN_P71 3 GND
JP84 GND_CAN_SHIELD_P84 GND
JP28 GND_ELECTRONICS_P28 GND

P1 GNDA
P2 SPARE_31C
P3 SPARE_32C
P4 SPARE_33C

J1A

1A	12V_FROM_MAIN_RELAY
2A	IN_CONSTANT_POWER
3A	OUT_FUEL_PUMP
4A	OUT_IDLE_CLOSE
5A	OUT_INJ_P03
6A	OUT_INJ_P04
7A	OUT_INJ_P05
8A	OUT_INJ_VVT_P07
9A	OUT_CHECK_ENGINE
10A	OEM_09
11A	OEM_10
12A	OEM_11
13A	OEM_12
14A	OEM_13
15A	GNDA_MAF_P14
16A	OEM_15
17A	IN_CRANK_P16
18A	GND
19A	IN_CAM_P17
20A	OEM_18
21A	OEM_19
22A	OEM_20
23A	OEM_21
24A	OUT_MAIN_RELAY
25A	GND_ELECTRONICS_P28
26A	GND
27A	OUT_IDLE_OPEN
28A	OEM_30
29A	OUT_INJ_P31
30A	OUT_INJ_P32
31A	OUT_INJ_P33
32A	OUT_INJ_P35
33A	OUT_EVAP
34A	OUT_O2_RELAY_P37

J1B

1B	IN_IGN_1
2B	IN_IGN_2
3B	IN_IGN_3
4B	IN_IGN_4
5B	IN_IGN_5
6B	IN_IGN_6
7B	IN_IGN_7
8B	IN_IGN_8
9B	OUT_O2_RELAY_P38
10B	GNDA_O2_P39
11B	GNDA_O2_P40
12B	IN_MAF
13B	IN_VSS_P42
14B	CRANK_RETURN_P43
15B	CAM_RETURN_P44
16B	GND_IGN_SHIELD_P45
17B	OUT_FUEL_CONSUMPTION
18B	OUT_TACH_P47
19B	OUT_AC_RELAY
20B	OEM_53
21B	OEM_57
22B	OEM_58
23B	+5VAS
24B	OEM_60
25B	OEM_61
26B	OEM_62

J2

1C	IN_VIGN
2C	OEM_63
3C	IN_AC_P64
4C	IN_AC_P65
5C	IN_IMMOBILIZER
6C	IN_KNOCK_P67
7C	IN_KNOCK_P68
8C	IN_KNOCK_P69
9C	IN_KNOCK_P70
10C	RETURN_P71
11C	OEM_72
12C	IN_TPS_P73
13C	OEM_74
14C	OEM_75
15C	OEM_76
16C	IN_IAT
17C	IN_CLT
18C	GND
19C	OEM_79
20C	OEM_80
21C	OEM_81
22C	OEM_82
23C	OEM_83
24C	GND_CAN_SHIELD_P84
25C	OEM_85
26C	GND
27C	OEM_86
28C	OEM_87
29C	OEM_88
30C	GNDA
31C	SPARE_31C
32C	SPARE_32C
33C	SPARE_33C
34C	GND

IN_CONSTANT_POWER
12V_FROM_MAIN_RELAY
IN_VIGN
+5VAS
GNDA
GND

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